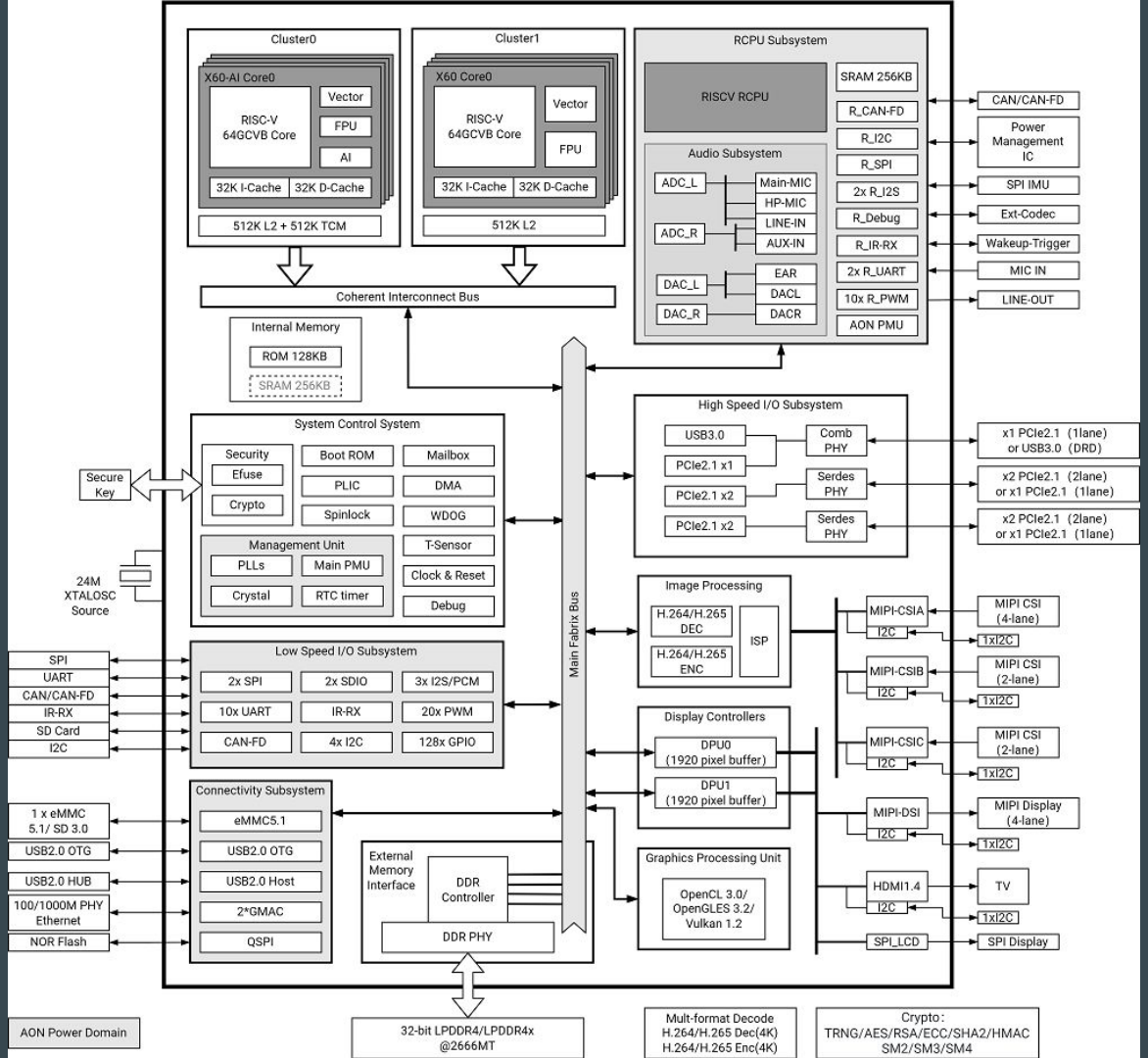
High Level Overview

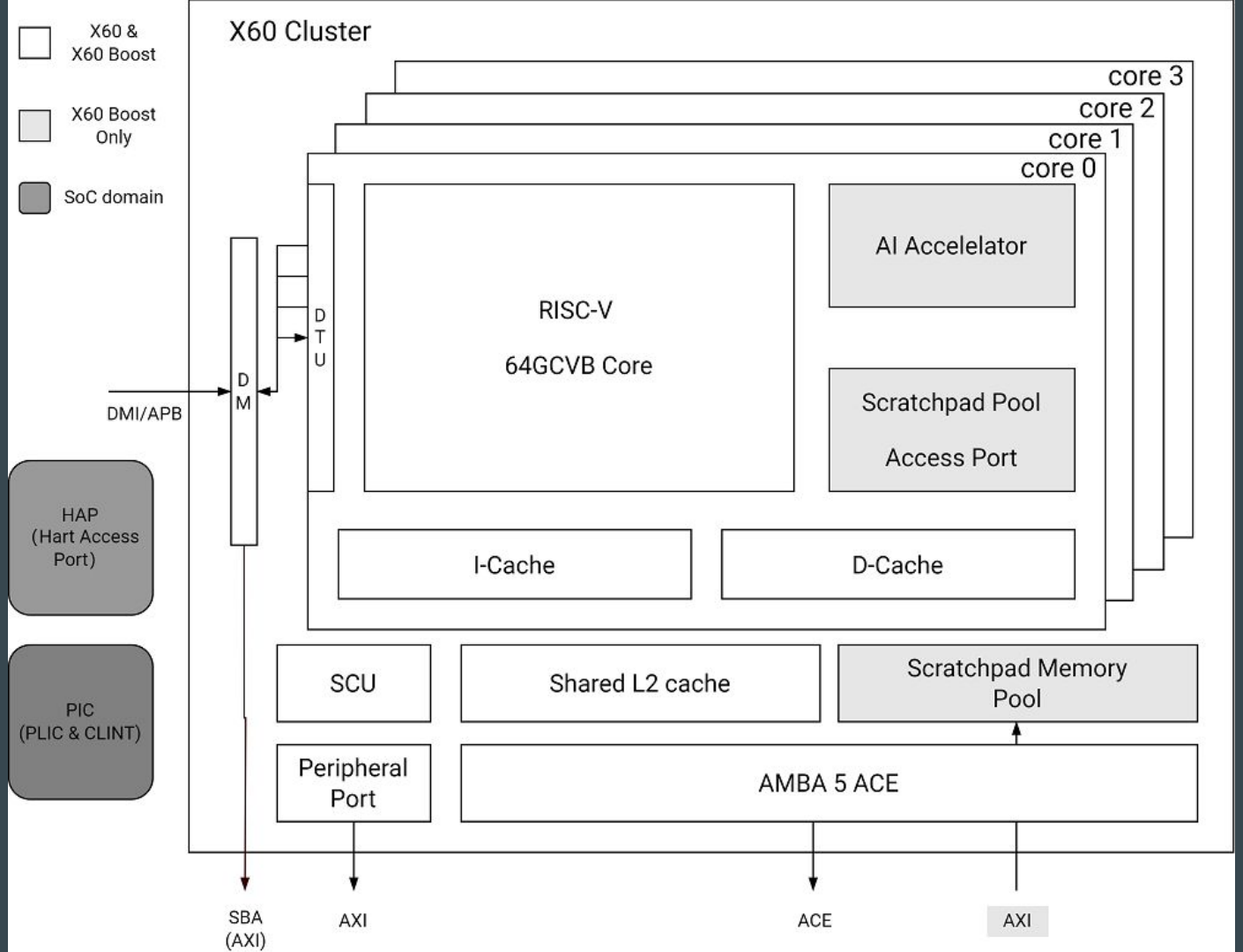
Octa-core RISC-V, dual-cluster asymmetric design

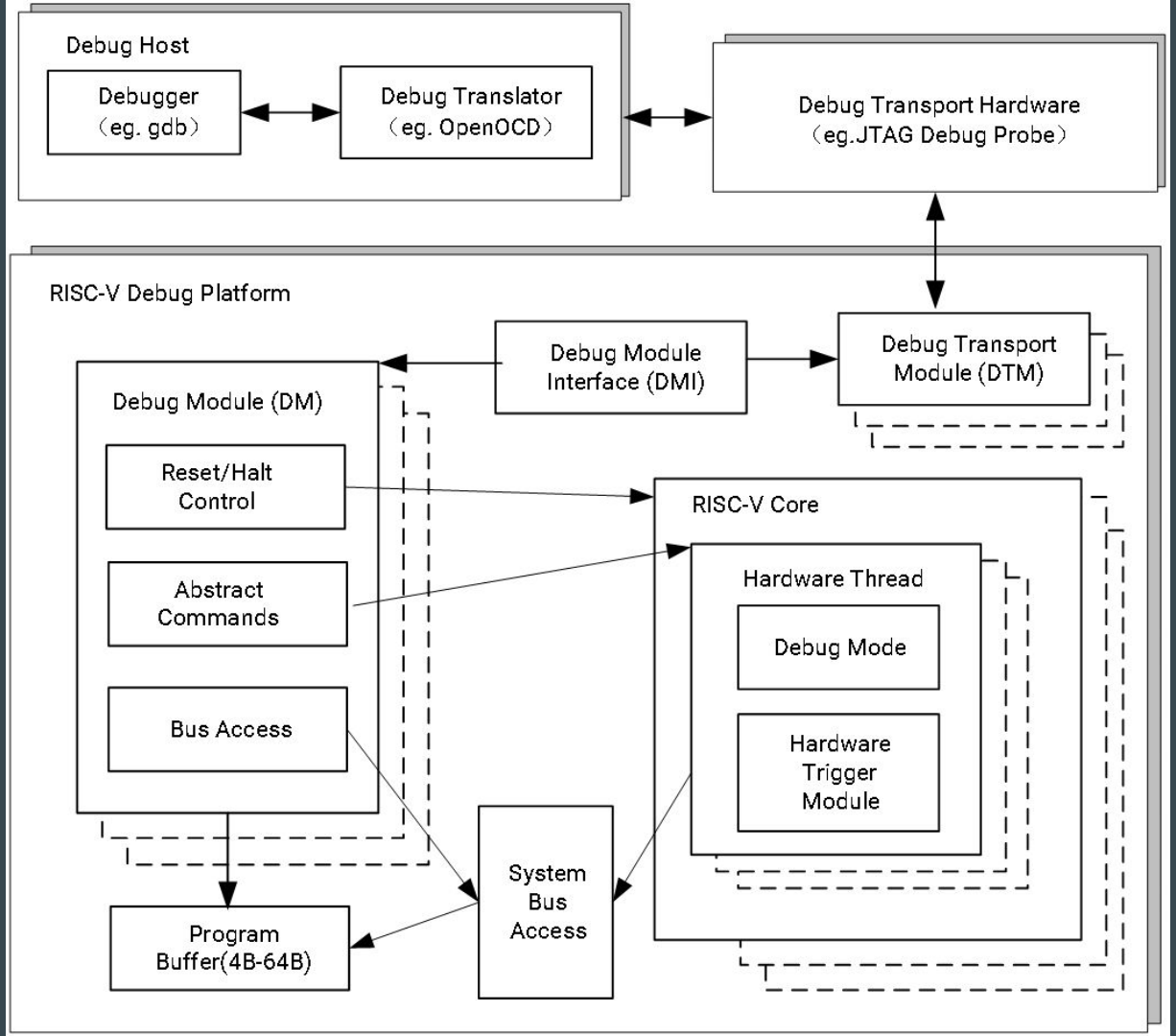
- Cluster 0:
  - AI-capable quad-core with 2.0 TOPS (Tera Operations Per Second)
  - 2.0 TOPS via custom RISC-V instructions — no discrete NPU, CPU-integrated
  - Supports TensorFlow Lite, ONNX Runtime (frameworks for AI models)
  - TCM (Tightly Coupled Memory – scratchpad memory on processor, zero latency + deterministic timing)
  - 512K L2
- Cluster 1: standard quad-core, 512K L2
- RVV 1.0: 256-bit vector processing (enables libjpeg-turbo)
- Dynamic Voltage and Frequency Scaling (DVFS) with voltage from 0.6V to 1.05V
  - Critical for conserving power in space

High Level Overview

- Watch Dog Design for each CPU
- Connectivity:
  - LPDDR4/4x up to 16GB (embedded applications)
  - eMMC 5.1, SD 3.0 (both for embedded storage, OS boot)
  - 3x PCIe Gen2 (connectivity with FPGA)
  - USB 3.0 (peripherals)
  - Dual GbE (good for benchmarking, not so much space application (ideally TSN))


Cluster0
X60-AI Core0
RISC-V 64GCVB Core
Vector
FPU
AI
32K I-Cache
32K D-Cache
512K L2 + 512K TCM
Cluster1
X60 Core0
RISC-V 64GCVB Core
Vector
FPU
32K I-Cache
32K D-Cache
512K L2
RCPU Subsystem
RISCV RCPU
SRAM 256KB
R_CAN-FD
R_I2C
R_SPI
2x R_I2S
R_Debug
R_IR-RX
2x R_UART
10x R_PWM
AON PMU
Audio Subsystem
ADC_L
ADC_R
Main-MIC
HP-MIC
LINE-IN
AUX-IN
DAC_L
DAC_R
EAR
DACL
DACR
CAN/CAN-FD
Power Management IC
SPI IMU
Ext-Codec
Wakeup-Trigger
MIC IN
LINE-OUT
Coherent Interconnect Bus
Internal Memory
ROM 128KB
SRAM 256KB
System Control System
Security
Efuse
Crypto
Boot ROM
PLIC
Spinlock
Mailbox
DMA
WDOG
T-Sensor
Clock & Reset
Debug
Management Unit
PLLs
Main PMU
Crystal
RTC timer
Secure Key
24M XTALOSC Source
Main Fabrix Bus
High Speed I/O Subsystem
USB3.0
PCIe2.1 x1
PCIe2.1 x2
PCIe2.1 x2
Comb PHY
Serdes PHY
Serdes PHY
x1 PCIe2.1 (1lane) or USB3.0 (DRD)
x2 PCIe2.1 (2lane) or x1 PCIe2.1 (1lane)
x2 PCIe2.1 (2lane) or x1 PCIe2.1 (1lane)
Image Processing
H.264/H.265 DEC
H.264/H.265 ENC
ISP
MIPI-CSIA
MIPI-CSIB
MIPI-CSIC
MIPI-DSI
HDMI1.4
SPI_LCD
I2C
1xI2C
MIPI CSI (4-lane)
MIPI CSI (2-lane)
MIPI CSI (2-lane)
MIPI Display (4-lane)
TV
SPI Display
Low Speed I/O Subsystem
2x SPI
2x SDIO
3x I2S/PCM
10x UART
IR-RX
20x PWM
CAN-FD
4x I2C
128x GPIO
SPI
UART
CAN/CAN-FD
IR-RX
SD Card
I2C
Display Controllers
DPU0 (1920 pixel buffer)
DPU1 (1920 pixel buffer)
Connectivity Subsystem
eMMC5.1
USB2.0 OTG
USB2.0 Host
2*GMAC
QSPI
1 x eMMC 5.1/ SD 3.0
USB2.0 OTG
USB2.0 HUB
100/1000M PHY Ethernet
NOR Flash
External Memory Interface
DDR Controller
DDR PHY
Graphics Processing Unit
OpenCL 3.0/ OpenGLES 3.2/ Vulkan 1.2
AON Power Domain
32-bit LPDDR4/LPDDR4x @2666MT
Mult-format Decode
H.264/H.265 Dec(4K)
H.264/H.265 Enc(4K)
Crypto:
TRNG/AES/RSA/ECC/SHA2/HMAC
SM2/SM3/SM4


X60 &
X60 Boost
X60 Boost
Only
SoC domain
X60 Cluster
core 3
core 2
core 1
core 0
AI Accelelator
RISC-V
64GCVB Core
Scratchpad Pool
Access Port
D
T
U
D
M
DMI/APB
HAP
(Hart Access
Port)
PIC
(PLIC & CLINT)
I-Cache
D-Cache
SCU
Shared L2 cache
Scratchpad Memory
Pool
Peripheral
Port
AMBA 5 ACE
SBA
(AXI)
AXI
ACE
AXI


Debug Host
Debugger
（eg. gdb）
Debug Translator
（eg. OpenOCD）
Debug Transport Hardware
（eg.JTAG Debug Probe）
RISC-V Debug Platform
Debug Module
Interface (DMI)
Debug Transport
Module (DTM)
Debug Module (DM)
Reset/Halt
Control
Abstract
Commands
Bus Access
RISC-V Core
Hardware Thread
Debug Mode
Hardware
Trigger
Module
System
Bus
Access
Program
Buffer(4B-64B)

RISC-V

RVA22: industry standard profile for 64 bit, mature OS

- Does not require RVV 1.0 vector processing
- SpacemiT K1 added SIMD capability to hardware-optimize AI processing and image processing (ffmpeg, libjpeg-turbo, OpenCV all support RVV 1.0)

Open-source

- Good toolchain support (compilers, debuggers, benchmarks)
- Flexible for low-power, AI adaptations/optimizations

Pros

Octa-core asymmetric design — useful for fault tolerance testing

- Cluster 0 vs Cluster 1 asymmetry gives architectural variety to test isolation and containment

DVFS with adaptive operating voltage from 0.6V to 1.05V

Linux support: easy to bring up, plenty of software ecosystem to work with

Controllers for GPIO, UART, I2C, SPI: supports rich peripheral set required for space

Ethernet interface: critical for reliable I/O

Cheap and accessible (relatively)

Cons

Zero radiation hardening — COTS chip, so real SEE/TID testing is not possible.

- Fault injection has to be fully software-simulated

Non-standard AI RISC-V extensions can be problematic for portability and integration

Memory Limits:

- No L3 cache
- Compared to Jetson Nano Orin
  - 1.5MB L2 cache vs the BPI 512K L2 cache
  - 128-bit DDR5 vs 32-bit DDR4
  - Banana-Pi has ~10x less memory bandwidth

China-sourced...